

Advanced Topics in Computer Graphics I

Glint Rendering



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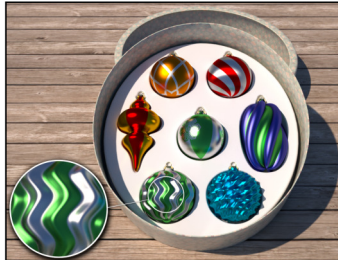
Institute of Computer Science II - Computer Graphics
University of Bonn

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Smooth Surface



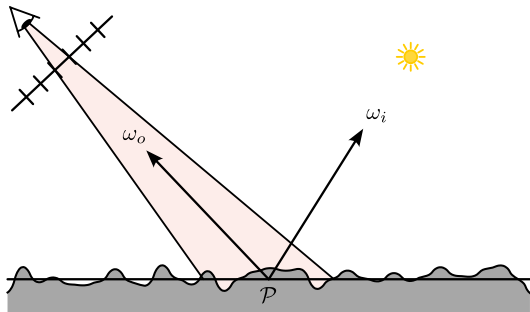
Glittering Surface



- Smooth surfaces exhibit synthetic/artificial appearance...



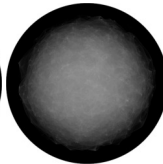
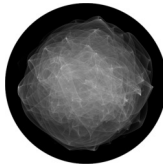
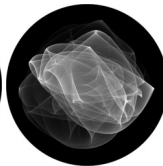
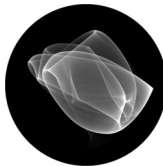
High Resolution Normal Maps



► Assumption: ω_i, ω_o constant for $\mathbf{x} \in \mathcal{P}$.

15×15

30×30



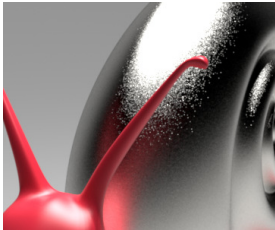
90×90

300×300

$\mathcal{P}$ -NDF Examples



Example



Actual Glints



Naive Sampling (4k spp)

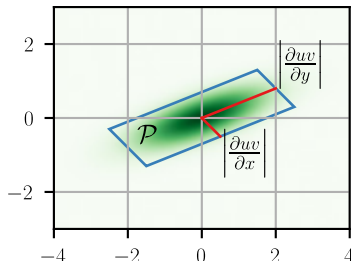


Single Pixel Closeup

- ▶ Naive sampling: $\mathbf{x} \in \mathcal{P}$ sampled randomly via sub-pixel offsets.
- ▶ Low probability of finding surface position $\mathbf{x}$ with correct normal orientation!



Pixel Footprint Shape



Uniform Box Function:

$$\int_{\mathcal{P}} p(\mathbf{x}) \, d\mathbf{x} = \int_A \frac{1}{|A|} \, d\mathbf{x} = 1$$

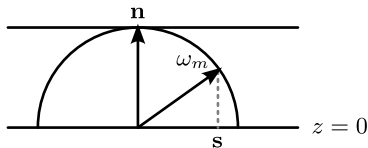
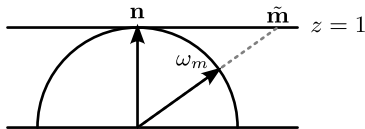
Gaussian Kernel:

$$\int_{\mathcal{P}} p(\mathbf{x}) \, d\mathbf{x} = \int_{\mathbb{R}^2} G_p(\mathbf{x}) \, d\mathbf{x} = 1$$

- Depending on the method, both conventions are used in practice...



- Directions from a 2D manifold in 3D
- Hemisphere is an impractical domain to work with



Slope Space Projection:

$$\tilde{\mathbf{m}} = (\tilde{m}_x, \tilde{m}_y) \in \mathbb{R}^2$$

$$\omega_m = (\tilde{m}_x, \tilde{m}_y, 1) / \sqrt{\tilde{m}_x^2 + \tilde{m}_y^2 + 1}$$

$$\left| \frac{\partial \omega_m}{\partial \tilde{\mathbf{m}}} \right| = \langle \mathbf{n}, \omega_m \rangle^4$$

Unit Disk Projection:

$$\mathbf{s} = (s_x, s_y) \in D_1$$

$$\|\mathbf{s}\| \leq 1$$

$$\omega_m = (s_x, s_y, \sqrt{1 - s_x^2 - s_y^2})$$

$$\left| \frac{\partial \omega_m}{\partial \mathbf{s}} \right| = \langle \mathbf{n}, \omega_m \rangle$$

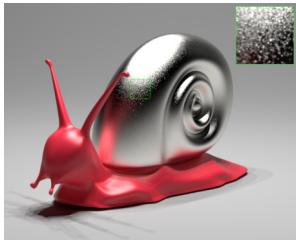
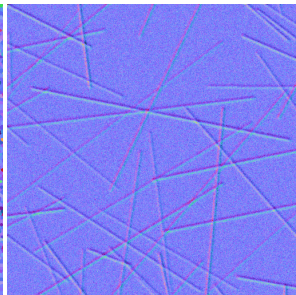
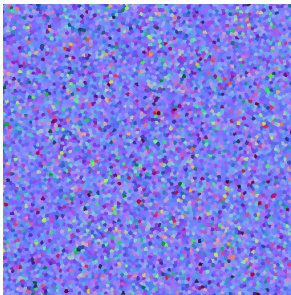
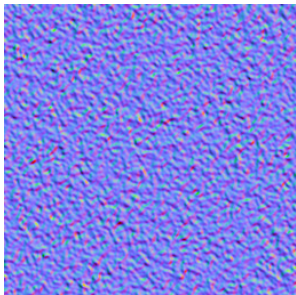


High Resolution Normal Maps for Glint Rendering

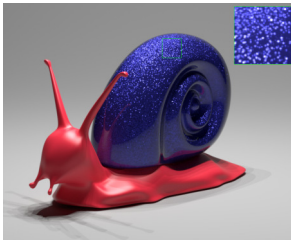




High Resolution Normal Maps



Isotropic Noise



Metallic Flakes



Scratches



- ▶ Integrate normal map (analytically) over pixel footprint $\mathcal{P}^1$.
- ▶ Represent position-normal distribution as Gaussian mixture².

¹L.-Q. Yan, M. Hašan, W. Jakob, et al. “Rendering glints on high-resolution normal-mapped specular surfaces”. In: *ACM Transactions on Graphics (TOG)* 33.4 (2014), pp. 1–9.

²L.-Q. Yan, M. Hašan, S. Marschner, et al. “Position-normal distributions for efficient rendering of specular microstructure”. In: *ACM Transactions on Graphics (TOG)* 35.4 (2016), pp. 1–9.



$$\begin{aligned} L_{o,\mathcal{P}}(\omega_o) &= \int_{\Omega_i} \int_{\mathcal{P}} \frac{F(\mathbf{x}, \langle \omega_i, \mathbf{h} \rangle) \cdot G(\mathbf{x}, \omega_i, \omega_o) \cdot D(\mathbf{x}, \mathbf{h})}{4 \langle \mathbf{n}, \omega_o \rangle \langle \mathbf{n}, \omega_i \rangle} d\mathbf{x} d\omega_i \\ &= \int_{\Omega_i} \frac{F(\langle \omega_i, \mathbf{h} \rangle) \cdot G(\omega_i, \omega_o)}{4 \langle \mathbf{n}, \omega_o \rangle \langle \mathbf{n}, \omega_i \rangle} \cdot \underbrace{\int_{\mathcal{P}} D(\mathbf{x}, \mathbf{h}) d\mathbf{x}}_{D_{\mathcal{P}}(\mathbf{h})} d\omega_i \end{aligned}$$

- ▶ $\mathcal{P}$ -NDF $D_{\mathcal{P}}(\mathbf{h})$.
- ▶ Gaussian footprint measure $G_p(\mathbf{x})$.

$$D_{\mathcal{P}}(\mathbf{h}) = \int_{\mathbb{R}^2} G_p(\mathbf{x}) \cdot D(\mathbf{x}, \mathbf{h}) d\mathbf{x}$$

³L.-Q. Yan, M. Hašan, W. Jakob, et al. “Rendering glints on high-resolution normal-mapped specular surfaces”. In: *ACM Transactions on Graphics (TOG)* 33.4 (2014), pp. 1–9.



- ▶ Unit Disk Projection $\mathbf{h} \rightarrow \mathbf{s} = (x_{\mathbf{h}}, y_{\mathbf{h}}) \in D_1$
- ▶ (Projected) normal map $\mathbf{n}(\mathbf{x}) \in D_1$
- ▶ Perfect mirror surface:

$$D_{\mathcal{P}}(\mathbf{h}) = \int G_p(\mathbf{x}) \cdot \delta(\mathbf{n}(\mathbf{x}) - \mathbf{s}) \, d\mathbf{x} = \sum_i \frac{G_p(\mathbf{x}_i)}{|\mathbf{n}'(\mathbf{x}_i)|}$$

- ▶ Problem if $\mathbf{n}'(\mathbf{x}_i) = 0$
- ▶ I.e. constant $\mathbf{n}(\mathbf{x})$



- Convolve with small Gaussian roughness $G_r(\mathbf{s})$:

$$\begin{aligned} D_{\mathcal{P}}(\mathbf{h}) &= \int_{\mathbb{R}^2} G_r(\mathbf{s} - \mathbf{s}') \int_{\mathbb{R}^2} G_p(\mathbf{x}) \cdot \delta(\mathbf{n}(\mathbf{x}) - \mathbf{s}') \, d\mathbf{x} \, d\mathbf{s}' \\ &= \int_{\mathbb{R}^2} G_p(\mathbf{x}) \int_{\mathbb{R}^2} G_r(\mathbf{s} - \mathbf{s}') \cdot \delta(\mathbf{n}(\mathbf{x}) - \mathbf{s}') \, d\mathbf{s}' \, d\mathbf{x} \\ &= \int_{\mathbb{R}^2} G_p(\mathbf{x}) \cdot G_r(\mathbf{n}(\mathbf{x}) - \mathbf{s}) \, d\mathbf{x} \\ &= \int_{\mathbb{R}^2} G_c[\mathcal{P}, \mathbf{s}](\mathbf{x}, \mathbf{n}(\mathbf{x})) \, d\mathbf{x} \end{aligned}$$

- Combined reconstruction kernel $G_c[\mathcal{P}, \mathbf{s}]$.
- Integration of a 4D position-normal Gaussian over 2D manifold $(\mathbf{x}, \mathbf{n}(\mathbf{x}))$!
- Given fine triangulation in $\mathbf{x} \rightarrow$ Analytic integration of 2D Gaussian over triangles.



Replace $G_r(\mathbf{n}(\mathbf{x}), \mathbf{s})$ with Gaussian mixture $\mathcal{N}(\mathbf{x}, \mathbf{s})$ with m elements.

$$\mathcal{N}(\mathbf{x}, \mathbf{s}) = \sum_{i=1}^m G_i(\mathbf{x}, \mathbf{s})$$

$$G_i(\mathbf{x}, \mathbf{s}) = c_i \cdot \exp\left(-\frac{1}{2}(\mathbf{v} - \mathbf{v}_i)^T \mathbf{\Sigma}_i^{-1}(\mathbf{v} - \mathbf{v}_i)\right),$$

where $\mathbf{v} = (\mathbf{x}, \mathbf{s})$ and $\mathbf{v}_i = (\mathbf{x}_i, \mathbf{s}_i)$.

$$D_{\mathcal{P}}(\mathbf{h}) = \sum_{i=1}^m \int_{\mathbb{R}^2} G_p(\mathbf{x}) \cdot G_i(\mathbf{x}, \mathbf{s}) \, d\mathbf{x}$$

- ▶ No more dependent arguments!
- ▶ Integral has closed-form solution.
- ▶ Note: m is large! ($\approx$ millions of elements)

⁴L.-Q. Yan, M. Hašan, S. Marschner, et al. "Position-normal distributions for efficient rendering of specular microstructure". In: *ACM Transactions on Graphics (TOG)* 35.4 (2016), pp. 1–9.



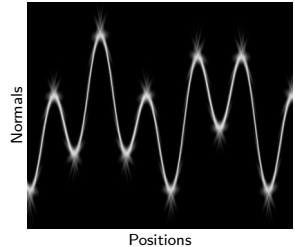
Avoid EM algorithms:

- ▶ Variable quality of learned distribution.
- ▶ Temporal coherence in animations.
- ▶ Target function (with Jacobians) explicitly known.

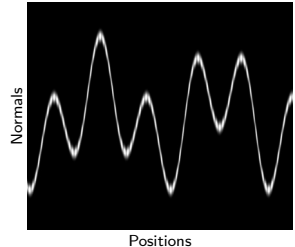
Instead:

- ▶ Instantiate Gaussians at fixed spatial intervals of length h (≈ 0.5 to 2 texels).
- ▶ Spatial std dev $\sigma_h := h / \sqrt{8 \log 2}$.
- ▶ Decay to half of peak value after distance $h/2$.
- ▶ Avoid overshooting and clumping.

Overshooting



Clumping



Flat Elements:

- Assume piecewise constant normal map.
- Independent spatial and directional components.
- Diagonal covariance matrix.

$$G_i(\mathbf{x}, \mathbf{s}) = c_i \cdot \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}_i\|^2}{2\sigma_h^2}\right) \cdot \exp\left(-\frac{\|\mathbf{s} - \mathbf{n}(\mathbf{x}_i)\|^2}{2\sigma_r^2}\right)$$

$$= c_i \cdot \exp\left(-\frac{1}{2}(\mathbf{v} - \mathbf{v}_i)^t \boldsymbol{\Sigma}_i^{-1} (\mathbf{v} - \mathbf{v}_i)\right)$$

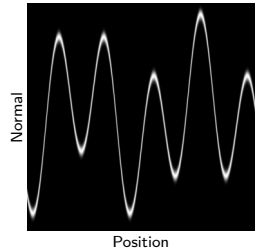
$$c_i := 1/h^2$$

$$\mathbf{v}_i := (\mathbf{x}_i, \mathbf{n}(\mathbf{x}_i))$$

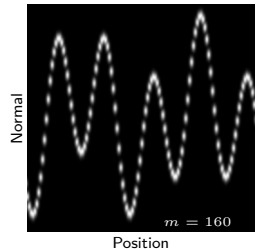
$$\boldsymbol{\Sigma}_i^{-1} := \text{diag}(\sigma_h^2, \sigma_h^2, \sigma_r^2, \sigma_r^2)^{-1}$$

- Problem: See illustration on the right.

Position-Normal Distribution



GMM with Flat Elements





Curved Elements:

- Assume piecewise linear normal map.

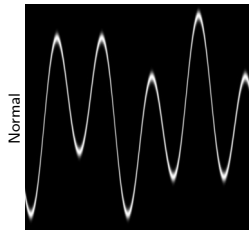
$$\mathbf{n}(\mathbf{x}) \approx \mathbf{n}(\mathbf{x}_i) + \mathbf{J} \cdot (\mathbf{x} - \mathbf{x}_i) \quad \mathbf{J} := \frac{\partial \mathbf{n}(\mathbf{x}_i)}{\partial \mathbf{x}_i}$$

- Expanding the definition of $G_i(\mathbf{x}, \mathbf{s})$ yields

$$\begin{aligned} G_i(\mathbf{x}, \mathbf{s}) &= c_i \cdot \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}_i\|^2}{2\sigma_h^2} - \frac{\|[\mathbf{s} - \mathbf{n}(\mathbf{x}_i)] - \mathbf{J}[\mathbf{x} - \mathbf{x}_i]\|^2}{2\sigma_r^2}\right) \\ &= c_i \cdot \exp\left(-\frac{1}{2}(\mathbf{v} - \mathbf{v}_i)^t \boldsymbol{\Sigma}_i^{-1} (\mathbf{v} - \mathbf{v}_i)\right) \end{aligned}$$

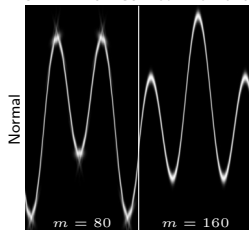
$$\boldsymbol{\Sigma}^{-1} := \frac{1}{\sigma_h^2} \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} + \frac{1}{\sigma_r^2} \begin{bmatrix} \mathbf{J}^t \mathbf{J} & -\mathbf{J}^t \\ -\mathbf{J} & \mathbf{I} \end{bmatrix}$$

Position-Normal Distribution



Position

GMM with Curved Elements



Position



- ▶ Given query footprint $G_p[\mathbf{x}]$ and query direction $\mathbf{s}[\mathbf{h}]$, only a small fraction of Gaussians G_i will contribute.
- ▶ Assume finite support for Gaussians $G_i(\mathbf{x}, \mathbf{s})$ (e.g. within 3σ).

Build bounding-volume hierarchy:

- ▶ Compute 4D bounding boxes in $(\mathbf{x}, \mathbf{s})$ -space.
- ▶ Compute top-down hierarchy via median-splits.

Improvements:

- ▶ Query direction $\mathbf{s}$ is a singular value:
 - ▶ Subdivide $\mathbf{s}$ dimensions into regular grid.
 - ▶ Build smaller hierarchy per cell.
- ▶ Gaussian evaluation is fast:
 - ▶ Stop build process early at e.g. 5 Gaussians per leaf node.

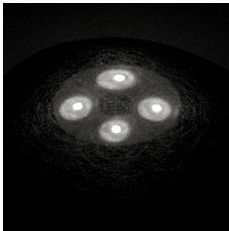


First bounce from camera: Importance-sample Gaussian mixture $\mathcal{N}(\mathbf{x}, \mathbf{s})$.

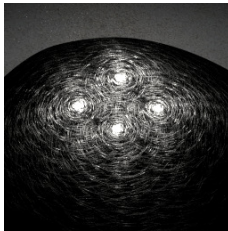
1. Select $G_i(\mathbf{x}, \mathbf{s})$ given $\mathcal{P}$
2. Then sample $\mathbf{s} \sim G_i(\mathbf{x}, \mathbf{s})$
3. Marginalize sampling probability for $\mathbf{s}$ over all candidate G_i for MIS.

Further bounces:

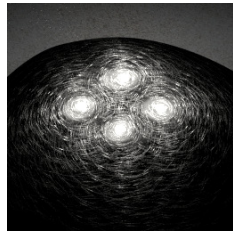
- Accurate $\mathcal{P}$ -NDF makes no difference anymore!
- Global $\mathcal{P}$ -NDF for sampling & evaluation.



$\mathcal{P}$ -NDF Sampling



Next Event Estimation



MIS Combination



Discrete Stochastic Microfacets





- ▶ N discrete microfacets distributed across the surface.

$$D(\mathbf{x}, \mathbf{h}) = \frac{1}{N} \sum_{k=1}^N \delta_A(\mathbf{x}, \mathbf{x}^{(k)}) \cdot \delta_{\Omega_h}(\mathbf{h}, \mathbf{h}^{(k)})$$

$$D(A, \Omega_h) = \int_A \int_{\Omega_h} D(\mathbf{x}, \mathbf{h}) \, d\mathbf{h} \, d\mathbf{x} = \frac{1}{N} \sum_{k=1}^N \mathbf{1}_A(\mathbf{x}^{(k)}) \cdot \mathbf{1}_{\Omega_h}(\mathbf{h}^{(k)})$$

- ▶ Finite area A instead of position $\mathbf{x}$
- ▶ Finite solid angle Ω_h instead of direction $\mathbf{h}$
- ▶ Count microfacets inside of A and Ω_h

⁵W. Jakob et al. "Discrete stochastic microfacet models". In: *ACM Transactions on Graphics (TOG)* 33.4 (2014), pp. 1–10.



“Regular” BRDF:

$$\rho(\mathbf{x}, \omega_i, \omega_o) = \frac{F(\langle \omega_i, \mathbf{h} \rangle) D(\mathbf{x}, \mathbf{h}) G(\omega_i, \omega_o)}{4 \langle \mathbf{n}, \omega_i \rangle \langle \mathbf{n}, \omega_o \rangle}$$

“Multiscale” BRDF:

$$\hat{\rho}(A, \omega_i, \Omega_o) = \frac{1}{a(A)\sigma(\Omega_o)} \int_A \int_{\Omega_o} \rho(\mathbf{x}, \omega_i, \omega_o) \, d\omega_o \, d\mathbf{x}$$

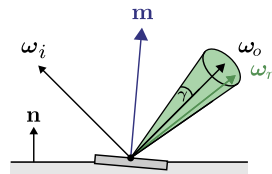
- ▶ Finite area A around $\mathbf{x}$
- ▶ Finite solid angle Ω_o around ω_o
- ▶ Single incident direction ω_i

$$\begin{aligned}
 \hat{\rho}(A, \omega_i, \Omega_o) &= \frac{1}{a(A)\sigma(\Omega_o)} \int_A \int_{\Omega_o} \rho(\mathbf{x}, \omega_i, \omega_o) \, d\omega_o \, d\mathbf{x} \\
 &= \frac{1}{a(A)\sigma(\Omega_o)} \int_A \int_{\Omega_h} \rho(\mathbf{x}, \omega_i, \omega_o) \cdot \left| \frac{\partial \omega_o}{\partial \mathbf{h}} \right| d\mathbf{h} \, d\mathbf{x} \\
 &= \frac{1}{a(A)\sigma(\Omega_o)} \int_A \int_{\Omega_h} \frac{F(\langle \omega_i, \mathbf{h} \rangle) D(\mathbf{x}, \mathbf{h}) G(\omega_i, \omega_o)}{4 \langle \mathbf{n}, \omega_i \rangle \langle \mathbf{n}, \omega_o \rangle} \cdot 4 \langle \mathbf{h}, \omega_o \rangle \, d\mathbf{h} \, d\mathbf{x} \\
 &= \frac{1}{Na(A)\sigma(\Omega_o)} \left[\sum_{k=1}^N \mathbf{1}_A(\mathbf{x}^{(k)}) \cdot \mathbf{1}_{\Omega_h}(\mathbf{h}^{(k)}) \right. \\
 &\quad \cdot \left. \frac{F(\langle \omega_i, \mathbf{h}^{(k)} \rangle) G(\omega_i, \omega_o^{(k)}) \langle \omega_i, \mathbf{h}^{(k)} \rangle}{\langle \omega_i, \mathbf{n} \rangle \langle \omega_i, \mathbf{n} \rangle} \right] \\
 &\approx \frac{1}{Na(A)\sigma(\Omega_o)} \left[\sum_{k=1}^N \mathbf{1}_A(\mathbf{x}^{(k)}) \cdot \mathbf{1}_{\Omega_h}(\mathbf{h}^{(k)}) \right] \cdot \frac{F(\langle \omega_i, \mathbf{h} \rangle) G(\omega_i, \omega_o) \langle \omega_i, \mathbf{h} \rangle}{\langle \omega_i, \mathbf{n} \rangle \langle \omega_i, \mathbf{n} \rangle}
 \end{aligned}$$

► Additional $4 \langle \omega_i, \mathbf{h} \rangle$ factor introduced due to change of variables $\omega_o \rightarrow \mathbf{h}$.

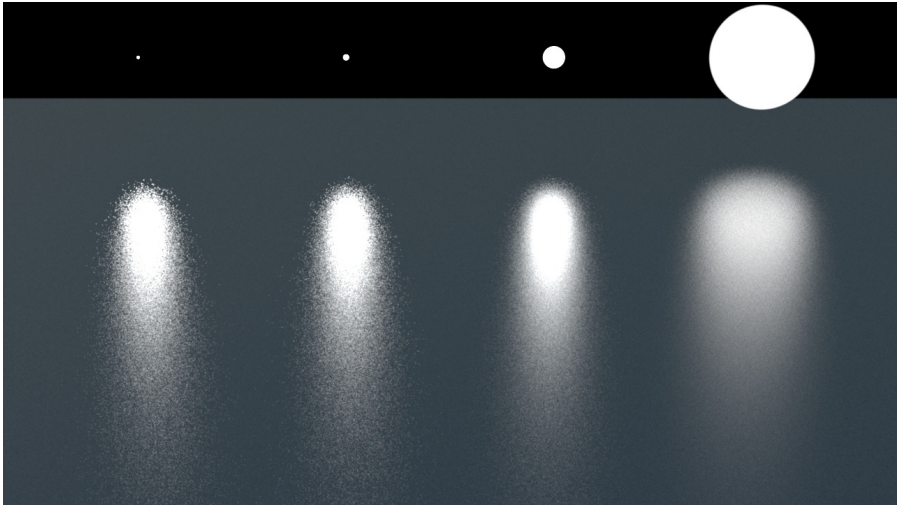


$$\begin{aligned}\hat{\rho}(A, \omega_i, \Omega_o) &= \frac{1}{Na(A)\sigma(\Omega_o)} \left[\sum_{k=1}^N \mathbf{1}_A(\mathbf{x}^{(k)}) \cdot \mathbf{1}_{\Omega_h}(\mathbf{h}^{(k)}) \right. \\ &\quad \cdot \left. \frac{F(\langle \omega_i, \mathbf{h}^{(k)} \rangle) G(\omega_i, \omega_o^{(k)}) \langle \omega_i, \mathbf{h}^{(k)} \rangle}{\langle \omega_i, \mathbf{n} \rangle \langle \omega_i, \mathbf{n} \rangle} \right] \\ &\approx \frac{1}{Na(A)\sigma(\Omega_o)} \left[\sum_{k=1}^N \mathbf{1}_A(\mathbf{x}^{(k)}) \cdot \mathbf{1}_{\Omega_h}(\mathbf{h}^{(k)}) \right] \\ &\quad \cdot \frac{F(\langle \omega_i, \mathbf{h} \rangle) G(\omega_i, \omega_o) \langle \omega_i, \mathbf{h} \rangle}{\langle \omega_i, \mathbf{n} \rangle \langle \omega_i, \mathbf{n} \rangle}\end{aligned}$$



(a) Planar view for one facet

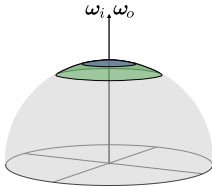
- Ω_o is a spherical cone with *small* angle γ around ω_o .
- Since $\sigma(\Omega_o)$ small, $\omega_o^{(k)} \approx \omega_o$, $\mathbf{h}^{(k)} \approx \mathbf{h}$ can be assumed.



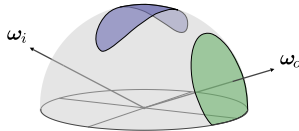
- ▶ Spherical cone angle γ controls the size of the assumed light source.
- ▶ Size of the light source affects the glint appearance!



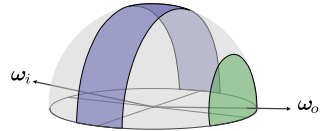
Spherical Conic Sections



(b) Spherical circle (0°)



(c) Spherical ellipse (140°)

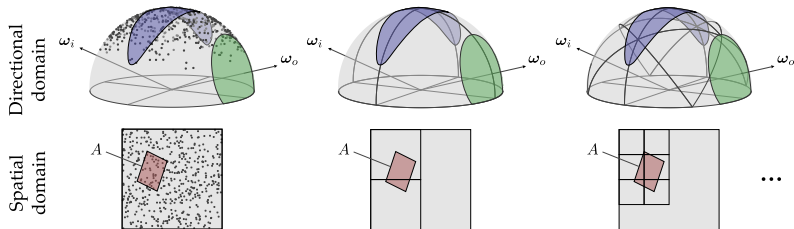


(d) Spherical hyperbola (170°)

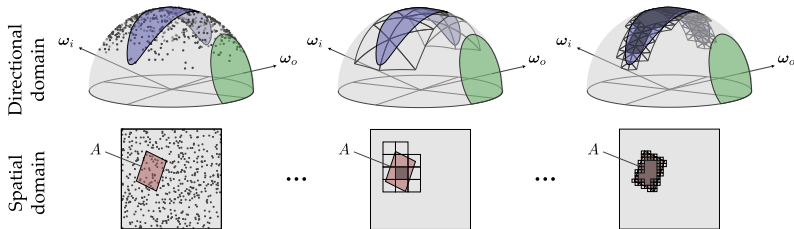
- Ω_o is a spherical cone around ω_o .
- Ω_h is a *second-order* spherical conic section around $\mathbf{h}$.

$$\begin{aligned}\mathbf{h}^{(k)} \in \Omega_h &\Leftrightarrow \omega_o^{(k)} \in \Omega_o \\ &\Leftrightarrow \langle \omega_o^{(k)}, \omega_o \rangle > \cos(\gamma) \\ &\Leftrightarrow 2\langle \mathbf{h}^{(k)}, \omega_i \rangle \langle \mathbf{h}^{(k)}, \omega_o \rangle - \langle \omega_i, \omega_o \rangle > \cos(\gamma) \\ &\Leftrightarrow \left(\mathbf{h}^{(k)} \right)^t \cdot \left[\mathbf{I} \cdot \frac{\cos(\gamma) + \langle \omega_i, \omega_o \rangle}{2} - \omega_i \cdot \omega_o^t \right] \cdot \mathbf{h}^{(k)} \leq 0\end{aligned}$$

Subdivide spatial and directional domains:



Final Subdivision:

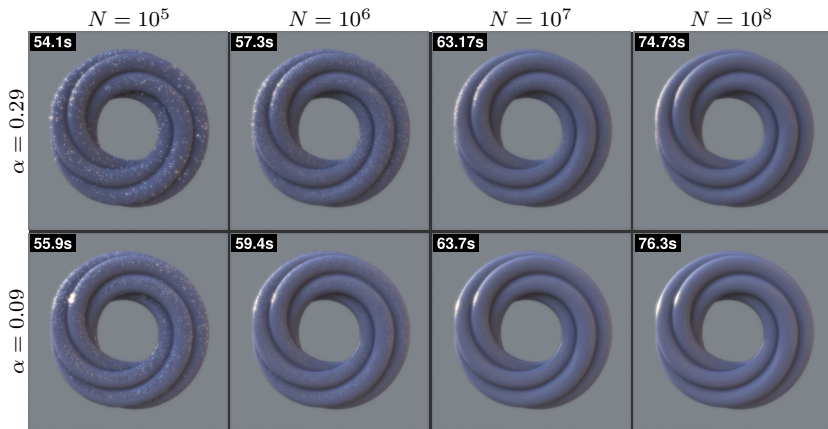


Algorithm Breadth-first search evaluation of the discrete NDF integral

```
1: function EVALNDFINTEGRAL( $N, A, \omega_i, \omega_o, \gamma$ )
2:   query  $\leftarrow A \times \text{CONICSECTION}(\omega_i, \omega_o, \gamma)$ 
3:   queue  $\leftarrow \{\text{root node of tree}\}$ 
4:   count  $\leftarrow 0$ 
5:   while queue  $\neq \emptyset$  do
6:     node  $\leftarrow \text{queue.POP}()$ 
7:     if node  $\cap$  query =  $\emptyset$  or |node| = 0 then
8:       pass
9:     else if node  $\subseteq$  query then
10:      count  $\leftarrow$  count + |node|
11:     else if Var[|node  $\cap$  query|] < ( $\varepsilon \cdot$  count)2 then
12:      count  $\leftarrow$  count +  $\mathbb{E}[| \text{node} \cap \text{query} |]$ 
13:     else
14:       for child  $\in$  node.SPLIT() do
15:         queue.PUSH(child)
16:       end for
17:     end if
18:   end while
19:   return count/ $N$ 
20: end function
```



Results



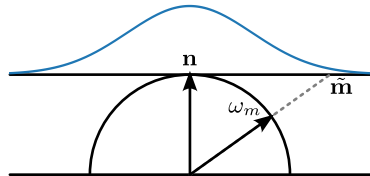
- Increasing N converges to smooth appearance.



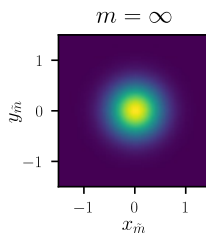
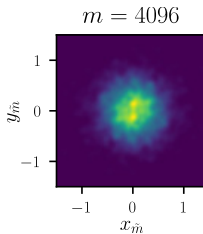
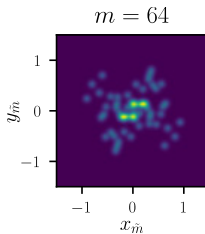
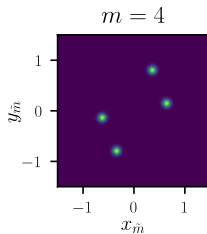
Procedural Patch NDFs for Real-time Rendering



- Normally distributed microfacets in slope space
- Small Gaussian roughness $\sigma_r = 0.02$
- 2D slope-space distribution function (SDF) $P^{22}(\tilde{\mathbf{m}})$



$$D_{\mathcal{P}}(\mathbf{h}) = \frac{P^{22}(\tilde{\mathbf{m}})}{\langle \mathbf{n}, \mathbf{h} \rangle^4}$$





Recap Beckmann Distribution

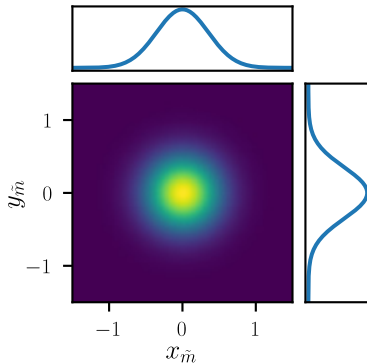
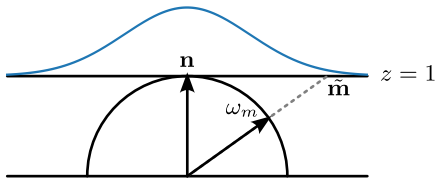


- ▶ Beckmann distribution $\Leftrightarrow$ Normal distribution in slope space.

$$\begin{aligned} D(\mathbf{h}) &= \frac{1}{\langle \mathbf{n}, \mathbf{h} \rangle^4} \cdot \frac{1}{\pi \alpha^2} \cdot \exp\left(\frac{\langle \mathbf{n}, \mathbf{h} \rangle^2 - 1}{\alpha^2}\right) \\ &= \left| \frac{\partial \mathbf{h}}{\partial \tilde{\mathbf{m}}} \right|^{-1} \cdot \frac{1}{2\pi \sigma^2} \cdot \exp\left(-\frac{\|\tilde{\mathbf{m}}\|^2}{2\sigma^2}\right) \end{aligned}$$

- ▶ Roughness $\alpha \Leftrightarrow$ Standard deviation σ

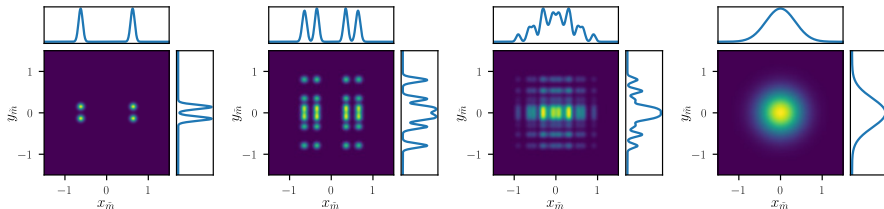
$$\alpha^2 = 2 \cdot \sigma^2$$





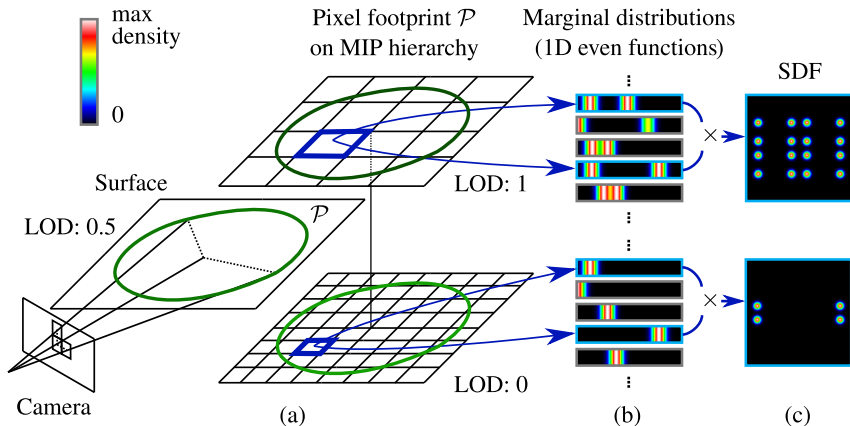
- ▶ Assume SDF $P^{22}(\tilde{\mathbf{m}})$ separable
- ▶ 1D marginal slope-space distribution functions $P^{2-}(x_{\tilde{m}})$ and $P^{-2}(y_{\tilde{m}})$

$$D_{\mathcal{P}}(\mathbf{h}) = \frac{P_{\mathcal{P}}^{22}(\tilde{\mathbf{m}})}{\langle \mathbf{n}, \mathbf{h} \rangle^4} = \frac{P^{2-}(x_{\tilde{m}}) \cdot P^{-2}(y_{\tilde{m}})}{\langle \mathbf{n}, \mathbf{h} \rangle^4}$$



⁶X. Chermain et al. "Procedural Physically based BRDF for Real-Time Rendering of Glints". In: *Computer Graphics Forum*. Vol. 39. 7. Wiley Online Library. 2020, pp. 243–253.

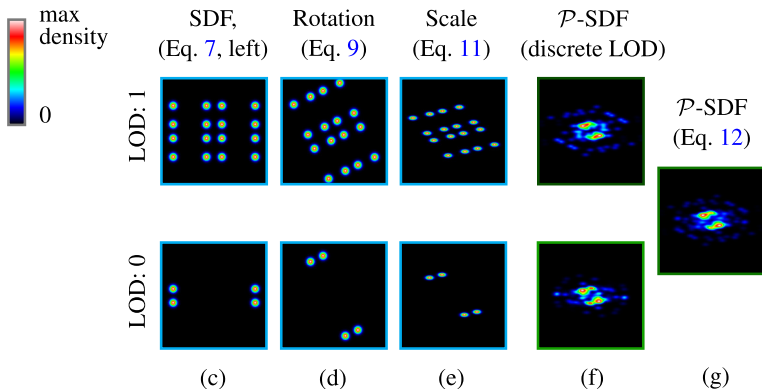
Query Dictionaries of Marginal Distributions



- Iterate over texels inside $\mathcal{P}$ on adjacent LOD levels.
- Query 1D marginal distributions.
- Decoded SDF from dictionary.



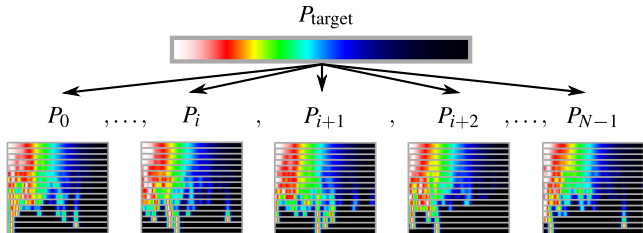
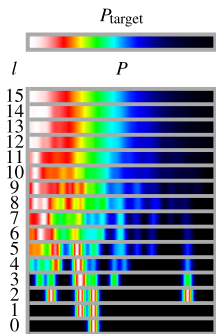
Reconstruction from Marginal Distributions



- c) Decoded SDF from dictionary.
- d) Random rotation to increase variety.
- e) Scale to account for roughness α .

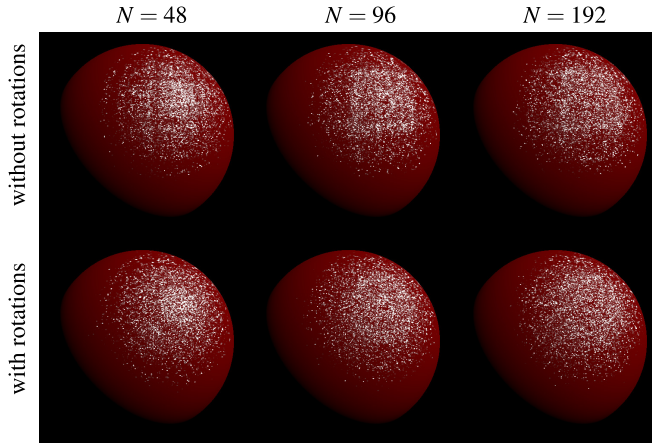
- f) $\mathcal{P}$ -SDF sum over texels on LOD level.
- g) Interpolation between LOD levels.

Multi-scale Marginal Distributions



- ▶ On LOD l , we have 2^l microfacets in 1D and 4^l microfacets in 2D.
- ▶ Each LOD step doubles the number of glints per dimension.
- ▶ $l \rightarrow \infty$ converge to smooth Beckmann distribution.
- ▶ Precompute dictionary of $N = 192$ LOD hierarchies.

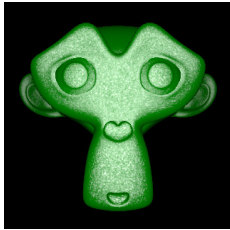
Number of Marginal Distributions



- ▶ Insufficient number of marginal distributions leads to visual artifacts.
- ▶ Applying random rotations helps.



Chermain et al. (2020)⁷



Yan et al. (2016)⁸



⁷X. Chermain et al. "Procedural Physically based BRDF for Real-Time Rendering of Glints". In: *Computer Graphics Forum*. Vol. 39. 7. Wiley Online Library. 2020, pp. 243–253.

⁸L.-Q. Yan, M. Hašan, S. Marschner, et al. "Position-normal distributions for efficient rendering of specular microstructure". In: *ACM Transactions on Graphics (TOG)* 35.4 (2016), pp. 1–9.



Real-time Rendering of Discrete Stochastic Microfacet with a Counting Model





Overview

Deliot & Belcour (2023)⁹Kneiphof & Klein (2024)¹⁰Kneiphof & Klein (2025)¹¹

DOI: 10.1111/vag.12006
High-Performance Graphics XE3
J. Wiley and C. Gribble
© 2014 John Wiley & Sons, Ltd.

COMPUTER GRAPHICS, *Journal*
Volume 42 (2018), Number 1
• 2018 • pp. 1 • 20

Partial Equations 2001-18, Class 1, Revised, and 15, White

Continuity: None

Geographical Symposium on Resilience 2002
B. Wang and A. Wilkin
Chengdu, China

COMPUTER GRAPHICS *Journal*
Volume 26 (2002), Number 1

Real-Time Rendering of Glinty Appearances using Distributed Binomial Laws on Anisotropic Grids

T. Delbecq and G. Belbin



Figure 1b We consider in real time gliders' appearances such as mass, size, point, or distinctive ornament.

Abstract
In this work, we consider real-time glinty materials caused by discrete flakes on the surface. To address this, one has to count the number of flakes reflecting the light towards the camera within every pixel covered by a given pixel footprint. To do so, we devise a counting method for arbitrary footprints that, unlike previous work, outputs the correct statistics. We combine this counting method with an anisotropic parameterization of the texture space that reduces the number of pixels falling inside a pixel footprint. This allows our method to run with both sub-pixel performance and $3.5 \times$ faster than the state-of-the-art.

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CCN Concepts
→ Computing methodologies → Modeling

1. Introduction

Thus, modeling different aspects of a greatly extends the volume of human-made material (car parts, molded plastics, ...) and natural ones (stones, water, snow, ...) adding to it a monitoring environment at very high scale. Indeed, such modeling requires high-resolution hydrological details, either due to complex boundaries (Fig. 14) or integrating high-resolution maps (SWAT 14). In contrast to other models, and due to limitations of physical surface mass fluxes, different aspects of what defines such models while at the same time preserving their appearance. This is a difficult task, and at each step, only a handful of models could be said to be able to do such modeling in practice (DRAFCAT 16).

Specifications for Real-Time. One major constraint is that the distribution of details must remain constant whenever the distance to the camera is, in a nutshell, if a particular plant exists at a given scale, it must continue until the camera moves closer or further. Real-time implementations often this constraint means coarse-to-fine coherence.

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The Computer Graphics and Info World & Info Lab

However, real-time methods still require that the underlying stochastic details converge towards a smooth, asymptotic model when the number of bonds tends to infinity.

INDEP as a Random Process. Using these qualifications, Zou and Kaplanian (2006) derived the fastest gluing shifter to be what the Normal Distribution Function (NDF) of a unimodal model (Wu06.007) is used as the success probability of multiple coin tosses. By connecting the number of trials to the number of faces within the die's footprint, they consider the randomness of gluing appearances. This process assumes both continuity as the shifting apparatus converges to the continuous normal distribution for an increasing number of trials. They further ensure the other side between top-levels by attaching random numbers on the object's surface in terms space.

However, this website reserves two drawbacks: First, it does not

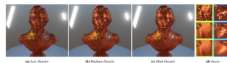


Figure 1: We compute the appearance of a glossy surface illuminated by two area light sources with equal overall intensity but different size in real time. (a) When the glints are sparsely distributed across the surface, linear microprisms are oriented to reflect light from the small light source (from the large light). The small light induces more pronounced glints than the large light for medium glint densities. (b) The appearance changes in the continuous range of glint densities. (c) The density of glints affects glint shapes to become as follows.

Abstract Many real-world materials are characterized by a diffuse appearance. Reproducing this effect in physically-based rendering is a challenging problem due to its diffuse nature, especially in real-time applications which require a consistently low runtime. Recent work focuses on diffuse appearance (transmission) in refraction and light sources only; for light sources like the sun and moon, rendering is a significant challenge. In this work, we extend previous work on rendering of light sources to include diffuse light sources. We derive an efficient algorithm for rendering along a straight ray by spatially constant diffuse areas (light) in real-time. In this end, we require an adequate estimate for the probability of a single ray intersecting to be correctly oriented for reflection from the source in the observer's A-gate estimate in a desired setting using linearly transformed cosine (LTC) for large light sources, and a linearly constant approximation of the normal distribution for small spherical light sources (light spheres). Compared to the resulting number of reflecting microsurfaces, we achieved a constant order on the bandwidth distribution. In this evaluation, we demonstrate the visual accuracy of our approach, which is already integrated into existing real-time rendering frameworks, e.g. light. If they already implement a rendering of light sources using LTC and a rendering model for given light sources, our pattern and color distribution, i.e. color distribution, is a good approximation for the rendering of light sources.

* Computing methodologies → Rendering; Refinement; modeling;

5. Introduction

The real world contains a vast number of materials with intricate light-scattering properties. Computer graphics researchers have been on their heels to find suitable digital representations that are cost effective and simultaneously produce accurate physically-based renderings for decades ([Santucci](#)). In particular, real-time applications,

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consistently every few milliseconds on cheap consumer hardware. Therefore, a full simulation of the light transport is often not feasible, requiring low-cost approximations that still produce results close to the ground truth. A common category of such complex materials are formed by those exhibiting glossy appearance, ranging from metallic car-paints over rough plastic material to literal glass, all kinds of decorative ornamentation and even granular solids.

Real-Time Image-based Lighting of Glints

Tom Kneipke¹ and Richard H.



Figure 1: Various glossy materials are rendered with our real-time image-based lighting approximation. From left to right: We show a woman wearing a glimmering dress, a car with metallic paint, and glossy component balls. Below are shown its ray-traced.

[illegible]

EXERCISES

1. Introduction

Since time immemorial, humankind has been attracted towards shiny and glittery objects [Lauff]. In this work, we are particularly interested in the sparkling effect of glass, which are usually the

smooth of microscopic mirror-like surface textures. Examples range from metallic car paints in the automotive industry to making sun-reflective glasses in the fashion industry, but also arises as a natural phenomenon in snow, stones, or when light is reflected from the

⁹T. Deliot and L. Belcour. "Real-Time Rendering of Glinty Appearances using Distributed Binomial Laws on Anisotropic Grids". In: *Computer Graphics Forum*. Vol. 42. 8. Wiley Online Library. 2023. e14866.

¹⁰T. Kneiphof and R. Klein. "Real-Time Rendering of Glints in the Presence of Area Lights". In: *Pacific Graphics Conference Papers and Posters*. Ed. by R. Chen et al. The Eurographics Association, 2024.

¹¹T. Kneiphof and R. Klein. “Real-time Image-based Lighting of Glints”. In: *Computer Graphics Forum* (2025).



Insert Conference Presentations here!

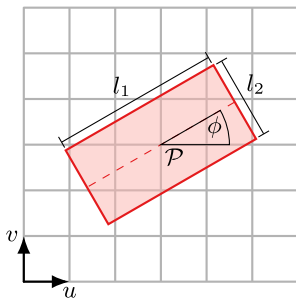


- ▶ Given probability for correct microfacet orientation p_{Ω_h} .
- ▶ Assume stratified distribution of microfacets, i.e. $N_{\mathcal{P}} = A_{\mathcal{P}} \cdot N_0$.
- ▶ Sample binomial distribution $b(N_{\mathcal{P}}, p_{\Omega_h})$ using uniform random numbers $\xi_k \in \mathcal{U}[0, 1]$.

$$b(N_{\mathcal{P}}, p_{\Omega_h}) = \sum_{k=1}^{N_{\mathcal{P}}} \chi(\xi_k < p_{\Omega_h})$$

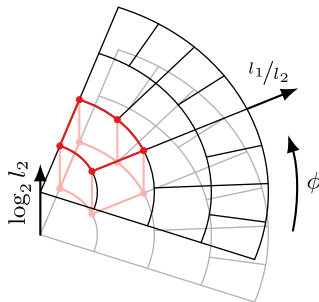
Problem(s):

- ▶ Temporal coherence
 - ▶ Want consistent results when the camera moves minimally or different subpixel locations are sampled for temporal anti-aliasing is used.
- ▶ *Variable* linear runtime $\mathcal{O}(N_{\mathcal{P}})$
 - ▶ $N_{\mathcal{P}}$ can become large.
 - ▶ Want constant runtime $\mathcal{O}(1)$.



Pixel parameters:

- ▶ Level of detail $\log l_2$.
- ▶ rotation in texture space ϕ .
- ▶ Anisotropic ratio l_1/l_2 .

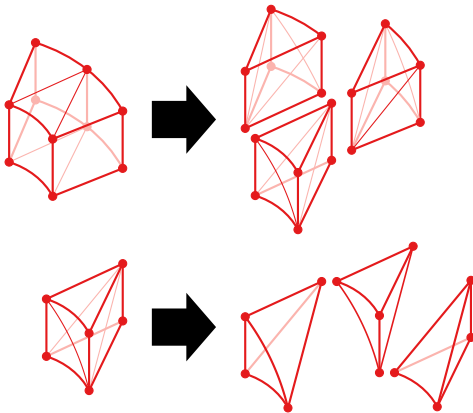


- ▶ Transformed into 3D radial grid
- ▶ Sample $b(N_P, p_{\Omega_h})$ at grid vertices
- ▶ Interpolate within pentagonal prism
- ▶ Special case $l_1/l_2 \rightarrow 0$: one face collapses into edge...



Problem:

- Interpolation over 10 pentagonal prism vertices is too much.



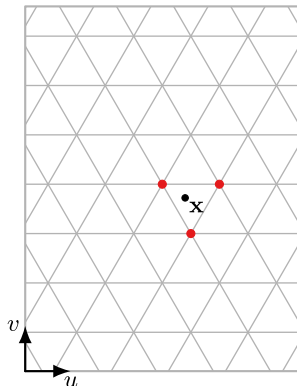
- A pentagonal prism is split into 3 prisms
- A prism is split into 3 tetrahedrons
- Only 4 adjacent vertices in interpolation.



- ▶ Skewed triangular grid for position $\mathbf{x}$ in uv space.
- ▶ 3 neighboring vertices instead of 4 vertices (rectilinear grid)
- ▶ Hexagonal shape on surface (almost round) instead of square.

- ▶ Basis vectors for skewed grid are x -axis and 60° rotated x -axis (in uv space).

$$T_{\text{skew} \leftarrow uv} = \begin{bmatrix} 1 & \cos(60^\circ) \\ 0 & \sin(60^\circ) \end{bmatrix}^{-1}$$

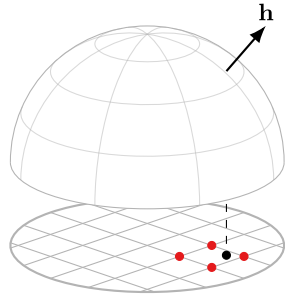




Halfway Grid



- ▶ Glints should change when light source moves.
- ▶ Microfacets have “some lifetime” in unit disk projection.
- ▶ Rectilinear grid with 4 vertices is not further optimized ...

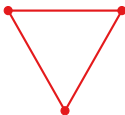




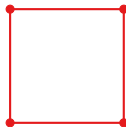
Pixel footprint $\mathcal{P}$



Position $\mathbf{x}$



Halfway vector $\mathbf{h}$



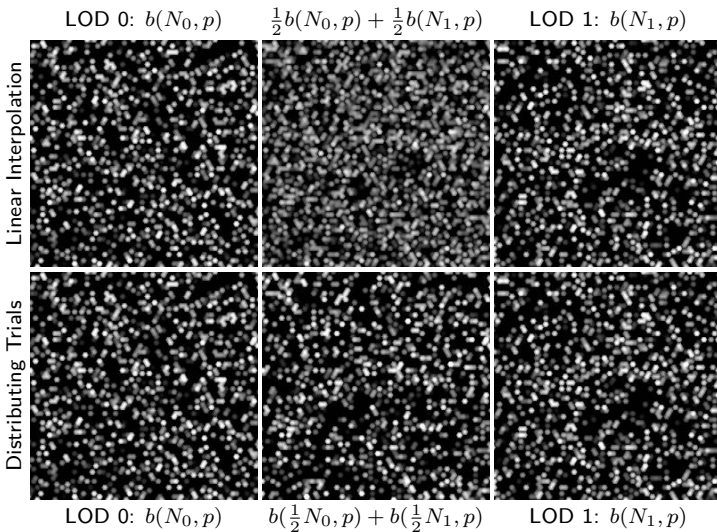
- Weighted sum over grid corners

$$b(N_{\mathcal{P}}, p_{\Omega_h}) = \sum_{i=1}^{4 \times 3 \times 4} w_i \cdot b_{\theta_i}(N_i, p_{\Omega_h})$$

- Distribute $N_{\mathcal{P}}$ over pixel footprint grid vertices $\rightarrow N_i$.
- Interpolate over position, halfway grid vertices $\rightarrow w_i$.



Distribution vs Interpolation?





- ▶ Assuming $N_{\mathcal{P}} \geq 1$: (Otherwise single Bernoulli trial with $p = N_{\mathcal{P}} \cdot p_{\Omega_h}$)
- ▶ Two uniform random numbers $\xi_{i,1}, \xi_{i,2} \in \mathcal{U}[0, 1]$ derived from seed θ_i

$$b_{\theta_i}(N_{\mathcal{P}}, p_{\Omega_h}) \approx \chi(\xi_{i,1} < P_{n \geq 1}) \cdot \text{clamp}_{[1, N_{\mathcal{P}}]}(\Phi^{-1}(\xi_{i,2}; \mu, \sigma^2))$$

- ▶ Bernoulli trial χ : Is there at least one positive outcome?

$$P_{n \geq 1} := 1 - (1 - p_{\Omega_h})^{N_{\mathcal{P}}}$$

- ▶ (Clamped) Gaussian with inverse CDF Φ^{-1}
- ▶ Mean, variance match underlying binomial distribution

$$\mu := N_{\mathcal{P}} \cdot p_{\Omega_h} \qquad \sigma^2 := N_{\mathcal{P}} \cdot p_{\Omega_h} \cdot (1 - p_{\Omega_h})$$

- ▶ **Note:** This works with positive real-valued $N_{\mathcal{P}}$, we don't need to round to integers!



- [1] X. Chermain, B. Sauvage, J.-M. Dischler, and C. Dachsbacher. “Procedural Physically based BRDF for Real-Time Rendering of Glints”. In: *Computer Graphics Forum*. Vol. 39. 7. Wiley Online Library. 2020, pp. 243–253.
- [2] T. Deliot and L. Belcour. “Real-Time Rendering of Glinty Appearances using Distributed Binomial Laws on Anisotropic Grids”. In: *Computer Graphics Forum*. Vol. 42. 8. Wiley Online Library. 2023, e14866.
- [3] W. Jakob, M. Hašan, L.-Q. Yan, J. Lawrence, R. Ramamoorthi, and S. Marschner. “Discrete stochastic microfacet models”. In: *ACM Transactions on Graphics (TOG)* 33.4 (2014), pp. 1–10.
- [4] T. Kneiphof and R. Klein. “Real-time Image-based Lighting of Glints”. In: *Computer Graphics Forum* (2025).
- [5] T. Kneiphof and R. Klein. “Real-Time Rendering of Glints in the Presence of Area Lights”. In: *Pacific Graphics Conference Papers and Posters*. Ed. by R. Chen, T. Ritschel, and E. Whiting. The Eurographics Association, 2024.
- [6] L.-Q. Yan, M. Hašan, W. Jakob, J. Lawrence, S. Marschner, and R. Ramamoorthi. “Rendering glints on high-resolution normal-mapped specular surfaces”. In: *ACM Transactions on Graphics (TOG)* 33.4 (2014), pp. 1–9.



- [7] L.-Q. Yan, M. Hašan, S. Marschner, and R. Ramamoorthi. “Position-normal distributions for efficient rendering of specular microstructure”. In: *ACM Transactions on Graphics (TOG)* 35.4 (2016), pp. 1–9.